FREQ SUITE

Backend Admin Guide

Operational reference for environment setup, device onboarding, authentication, WebSocket behavior, analytics, and reporting in the Freq backend service.

Service Version	Document Date	Contents
Freq Backend v1.0	March 2026	Configuration, routes, roles, provisioning, live flow, reports, and troubleshooting.

1. Service Overview

The backend provides REST APIs, dashboard authentication, device provisioning, dashboard WebSocket events, PDF reporting, and frequency analytics built from ingested device data.

Core backend responsibilities:

- Register and authenticate devices
- Maintain live dashboard and device session state
- Store commands, crashes, statistics, and frequency segments
- Generate licences and PDF reports

2. Authentication and Roles

Area	Description
Dashboard login	Uses `/api/auth/login` and returns a JWT token for frontend sessions.
Admin role	Can view all pages and send commands to devices.
User role	Can view all data but cannot dispatch commands.
Device login	Returns the device identifier and WebSocket address used for the live tunnel.

The backend currently supports local dashboard authentication, while remaining structured for future LDAP or Active Directory integration.

3. Device Provisioning and Live Flow

Devices use the open provisioning flow intended for the internal cruise-ship environment. Provisioning creates or updates device registration data and enables the device to proceed into login and WebSocket authentication.

Step	Result
`/api/devices/provision`	Registers the device and returns the provisioning result.
`/api/devices/login`	Returns device login result and `WSS_URL` for live connection.
WebSocket `auth`	Marks the device online and starts live dashboard updates.
Device logout or disconnect	Marks the device offline and broadcasts the session change.

4. Analytics and Reporting

Frequency data is ingested as segments and analyzed into fixed sessions. The backend correlates crashes against these sessions and surfaces summary values to both the frontend and the generated device report PDF.

- Total observed time
- Total fixed time
- Fixed session count
- Crashes during fixed sessions
- Crashes outside fixed sessions
- Top fixed frequencies for small and big cores

Reports are generated one device per page and include a frequency summary, fixed-session rows, and crash history.

5. Quick Reference

Administrator action	Result
Configure `.env` and restart the backend	Applies updated secrets, public WebSocket host, and runtime settings.
Create a dashboard user	Adds an `admin` or `user` account for frontend access.
Run the demo seed route	Populates groups, devices, licences, commands, crashes, and frequency data.
Review one-device analytics	Confirms fixed sessions and crash correlation are computed correctly.
Generate a report	Produces a multi-page PDF with one page per selected device.

End of guide.